

Task 8

<https://edaplayground.com/x/6fN4>

if $ab = 10$
 $f = 0 + 0 + 0 + 0 + 0$
 $f = 0$
 if $ab = 11$
 $f = 1$

Extension

a	b	c	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

$F = \sum m(3, 5, 6, 7)$

select a as select line
 if $a = 1$ $F = b + c$
 if $a = 0$ $F = bc$

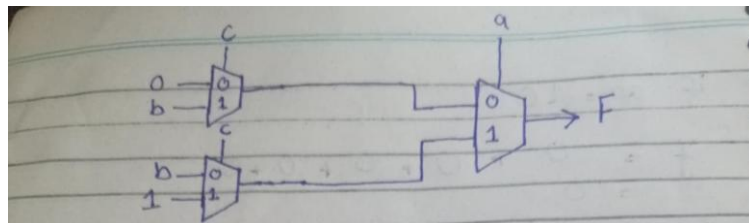
Task 8

$F = \sum m(1, 2, 3, 6, 7)$

a	b	c	F
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

4-1 mux

$f = a'b'c + a'bc' + a'bc$
 if $ab = 00$ $f = c + 0 + 0 + 0 + 0$
 $f = c$
 if $ab = 01$ $f = 0 + c' + c + 0 + 0$
 $f = 1$



Log

Share

Chronologic VCS simulator copyright 1991-2025

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Compiler version X-2025.06-SP1_Full64; Runtime version X-2025.06-SP1_Full64; Jul 20 11:14 2026

a b c | majority3 m12367 | Result

```

0 0 0 | 0 0 |
0 0 1 | 0 1 |
0 1 0 | 0 1 |
0 1 1 | 1 1 |
1 0 0 | 0 0 |
1 0 1 | 1 0 |
1 1 0 | 1 1 |
1 1 1 | 1 1 |

```

\$finish called from file "testbench.sv", line 40.

\$finish at simulation time 45

VCS Simulation Report

Time: 45 ns

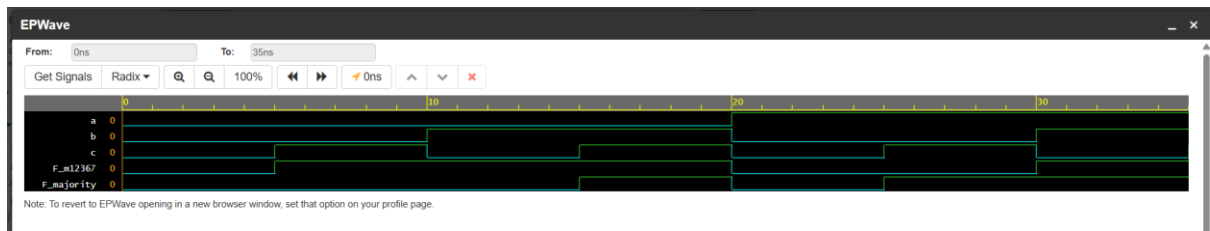
CPU Time: 0.520 seconds;

Data structure size: 0.0Mb

Mon Jul 20 11:14:00 2026

Done

Terminal Output



Output Waveform